FN585 Financial Modelling & Dealing

Week 3 (lecture)

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Time Series Data

- Time series data is data collected for a single entity at multiple points in time
  - Profits of a particular firm
  - A country's quarterly GDP
  - A region's monthly unemployment rate
  - Daily Covid-19 infections
- Univariate series data on $Y : \{Y_t; t = 1, 2, \ldots, T\}$
- Time series data can be used for causal analysis (but now, dynamic, i.e. over time) and prediction (now called forecasting)

Time Series Data

- So far, we have considered cross-sectional data: the subscript i relates to discrete units (individuals, households, regions and so on). The ordering of the observations in the sample along i doesn't matter.
- When we have time series data, the time dimension imposes a natural ordering. The total number of observations is denoted by T.

Time Series Data

- Another feature of time series is that the process we are interested in is typically continuous, not discrete. In standard econometrics, the data for a time series variable are converted into discrete form by taking records at regular intervals (think, for instance, of the observation of the interest rate r_t - it changes continuously over time although we only take records at regular intervals).

Time Series Data

- **Forecasting** (SW Ch. 15)
- **Estimation of** *dynamic* **causal effects** (SW Ch. 16)
  - If the Fed increases the Federal Funds rate now, what will be the effect on the rates of inflation and unemployment in 3 months? in 12 months?
  - What is the effect **over time** on cigarette consumption of a hike in the cigarette tax?
- **Modeling risks**, which is used in financial markets (one aspect of this, modeling changing variances and "volatility clustering," is discussed in SW Ch. 17)

US Quarterly GDP

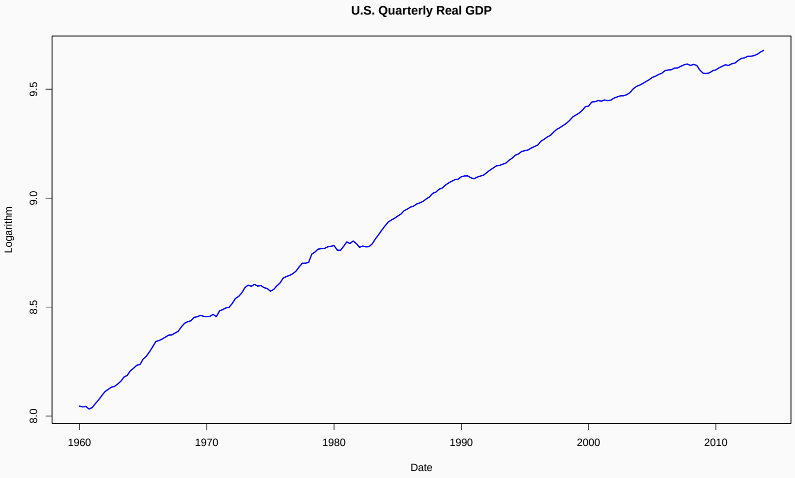
```r
usmac_qt <- read_xlsx("us_macro_quarterly.xlsx",sheet = 1,
                      col_types = c("text", rep("numeric", 9)))
```

```
## New names:
## * `` -> `...1`
```

```r
# Fix format of date
usmac_qt$...1 <- as.yearqtr(usmac_qt$...1, format = "%Y:0%q")
# Relabel column names in dataframe
colnames(usmac_qt) <- c("Date", "GDPC96", "JAPAN_IP", "PCECTPI",
"GS10", "GS1", "TB3MS", "UNRATE", "EXUSUK",
"CPIAUCSL")
# GDP series as xts object
GDP <- xts(usmac_qt$GDPC96, usmac_qt$Date)["1960::2013"]
# GDP growth series as xts object
GDPGrowth <- xts(400 * log(GDP/lag(GDP)))
```

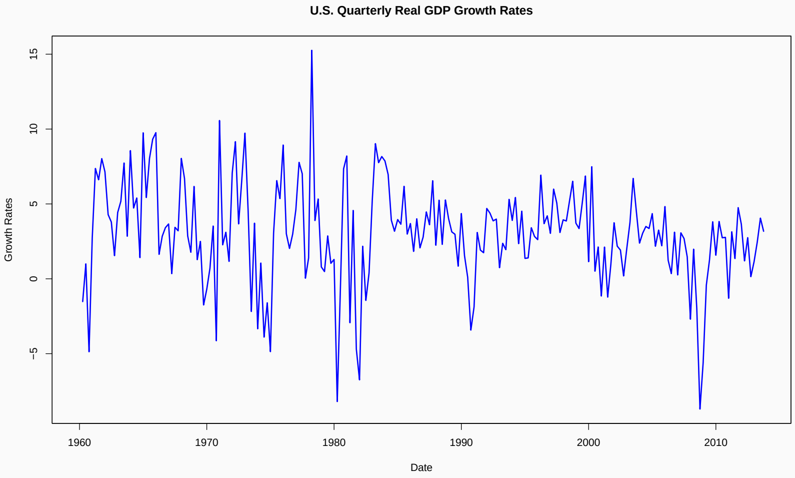
US Quarterly GDP

```
plot(log(as.zoo(GDP)),col = "blue",lwd = 2,ylab = "Logarithm",
     xlab = "Date", main = "U.S. Quarterly Real GDP")
```



US Quarterly GDP Growth Rates

```
plot(as.zoo(GDPGrowth), col = "blue", lwd = 2, ylab = "Growth Rates",
     xlab = "Date",  main = "U.S. Quarterly Real GDP Growth Rates")
```



Time series models - some definitions

- We talk about the variable X taking the form of a sequence $\{\cdots, X_1, \cdots, X_T, \cdots\}$, where individual observations are themselves random variables with their own distribution.
- As a consequence of the continuous nature of the observations, many regressors in time series exhibit what is known as persistence or dependence, which is successive observations are generally correlated.

some definitions

Lags and leads

Y_{t-1} : 1st lag of Y_t

Y_{t-j} : jth lag of Y_t

Y_{t+1} : 1st lead of Y_t

Y_{t+j} : jth lead of Y_t

some definitions

First difference

$\Delta Y_t = Y_t - Y_{t-1}$

The percentage chnage of Y_t between periods $t - 1$ and t is approximately $100\Delta \log Y_t$, where this approximation is most accurate when the percentage change is small.

Example: **Quarterly rate of growth of U.S. GDP at an annual rate**

GDP = Real GDP in the US (Billions of \$2009)

- **GDP in the fourth quarter of 2016 (2016:Q4) = 16851**
- **GDP in the first quarter of 2017 (2017:Q1) = 16903**
- **Percentage change in GDP, 2016:Q4 to 2017:Q1**

$$= 100 \times \left(\frac{16903 - 16851}{16851} \right) = 0.31\%$$

some definitions

- **Percentage change in GDP, 2012:Q1 to 2012:Q2, at an annual rate**

$$= 4 \times 0.31\% = \mathbf{1.23\%} \approx \mathbf{1.2\%} \text{ (percent per year)}$$

- **Using the logarithmic approximation to percent changes yields**

$$4 \times 100 \times [\log(16903) - \log(16851)] = \mathbf{1.232\%}$$

Autocovariance

Population Autocovariance

- Autocovariance is the covariance b/w a random variable Y in two periods
- The 1st autocovariance of Y_t is $Cov(Y_t, Y_{t-1})$; the jth autocovariance is $Cov(Y_t, Y_{t-j})$
- The jth autocorrelation is:

$$\rho_j = \frac{Cov(Y_t, Y_{t-j})}{\sqrt{Var(Y_t)}\sqrt{Var(Y_{t-j})}}$$

- Autocorrelations also called serial correlations - We refer to $Cov(Y_t, Y_{t-j})$ and ρ_j as a function of lag lenght j as autocovariance and autocorrelation functions

Sample Autocovariance

- The analogue principle suggests the following estimators have good statistical properties:

$$\widehat{\text{Cov}\left(Y_t, Y_{t-j}\right)} = \frac{1}{T} \sum_{t=j+1}^{T} \left(Y_t - \bar{Y}_{j+1:T}\right) \left(Y_{t-j} - \bar{Y}_{1:T-j}\right)$$

$$\hat{\rho}_j = \widehat{\text{Cov}\left(Y_t, Y_{t-j}\right)} / \widehat{\text{Var}\left(Y_t\right)}$$

where $\bar{Y}_{j+1:T}$ is sample average of Y from observations $t = j+1, \ldots, T$; $\widehat{\text{Var}\left(Y_t\right)}$ is sample variance of Y

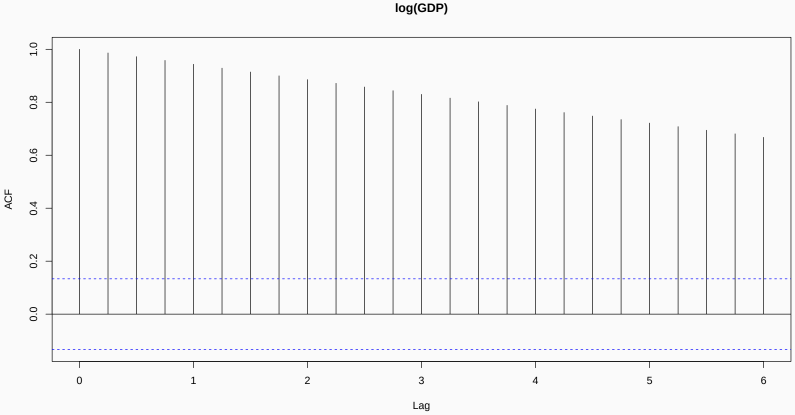
The Autocovariance Function in R

```r
#ACF of US quarterly GDP
acf <- acf(na.omit(log(GDP)), lag.max = 8, plot = FALSE)
acf
```

```
##
## Autocorrelations of series 'na.omit(log(GDP))', by lag
##
##  0.00  0.25  0.50  0.75  1.00  1.25  1.50  1.75  2.00
## 1.000 0.986 0.972 0.958 0.944 0.929 0.914 0.900 0.886
```

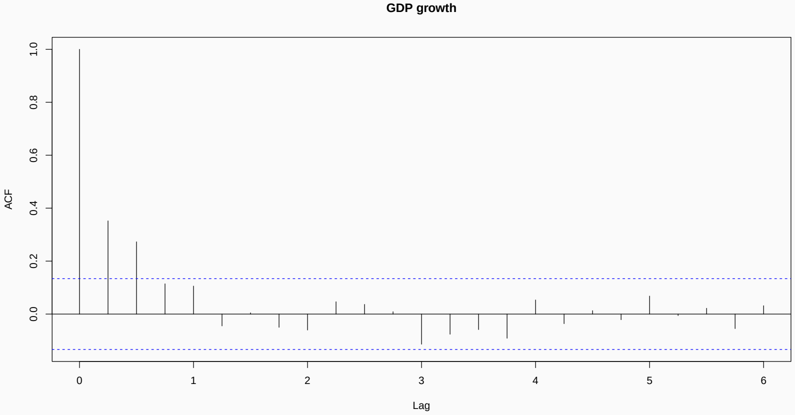
The Autocovariance Function Plot in R

```r
#ACF of US quarterly GDP
acf <- acf(na.omit(log(GDP)), lag.max = 24, plot = FALSE)
plot(acf,main="log(GDP)")
```

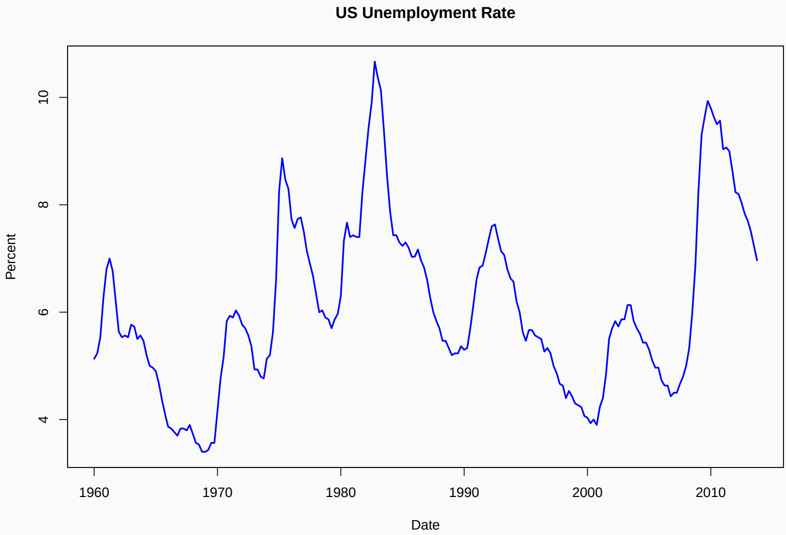


The Autocovariance Function Plot in R

```r
# ACF of US quarterly GDP growth
acf <- acf(na.omit(GDPGrowth), lag.max = 24, plot = FALSE)
plot(acf, main = "GDP growth")
```



Some examples of economic time series



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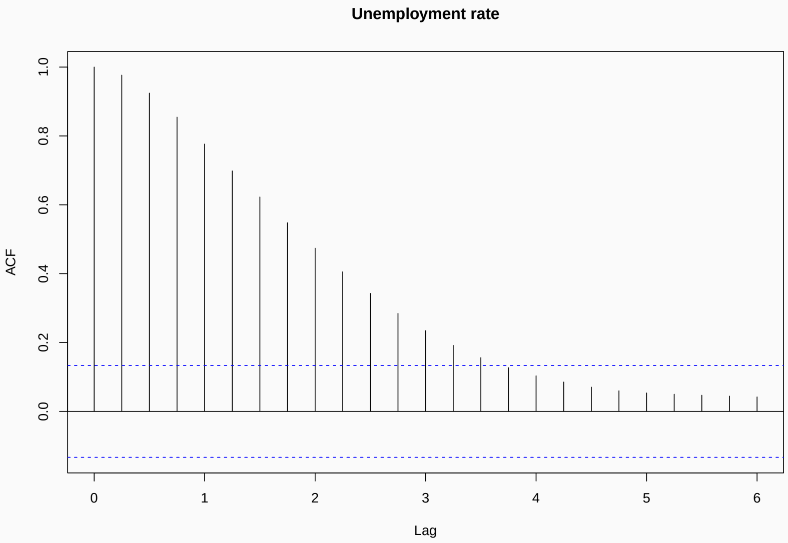
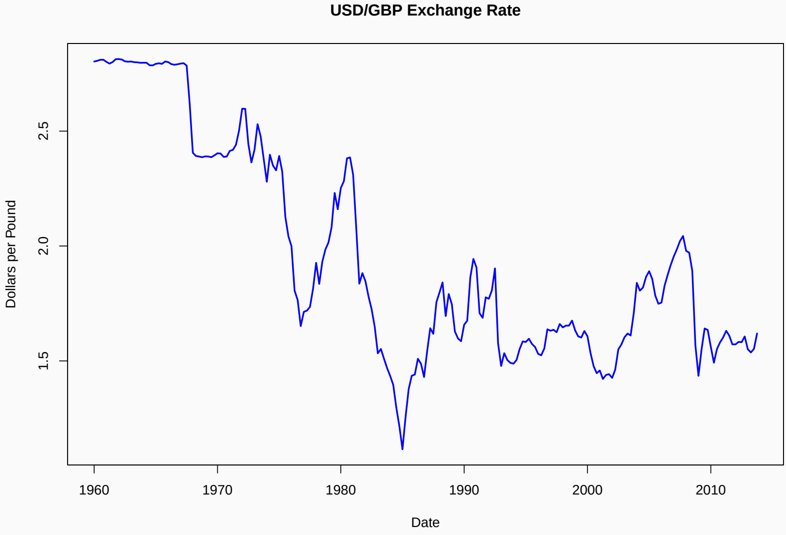
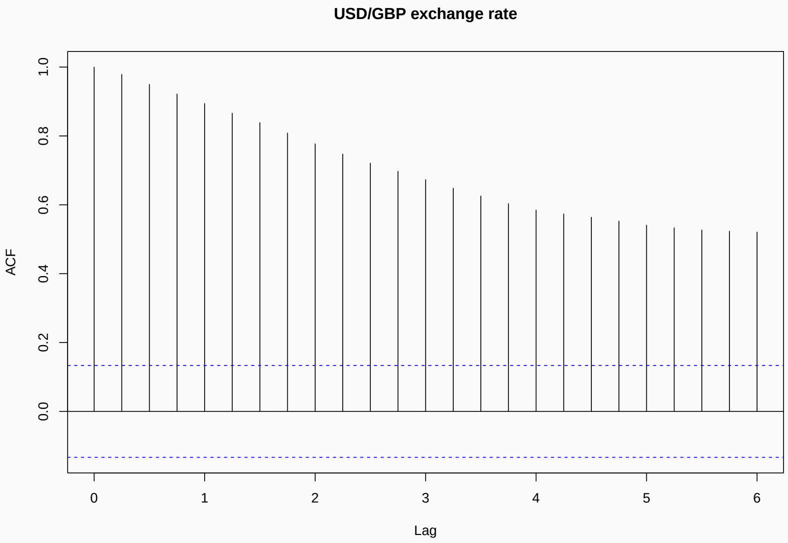


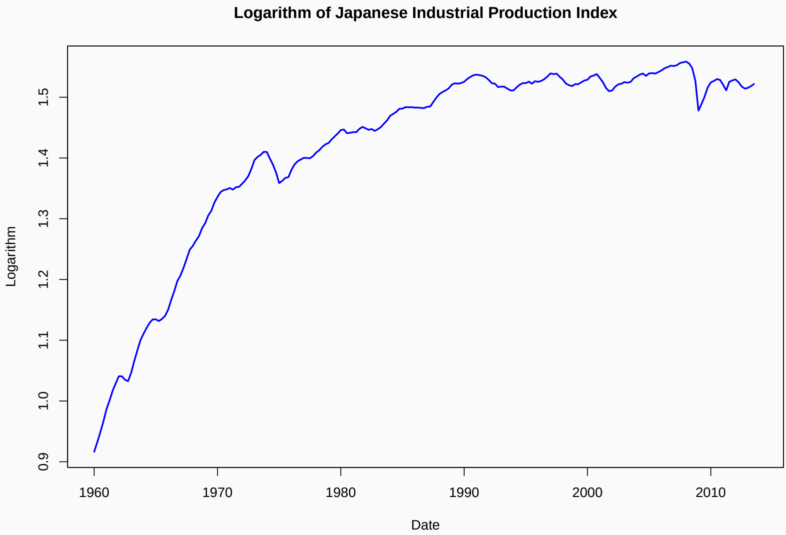
Some examples of economic time series



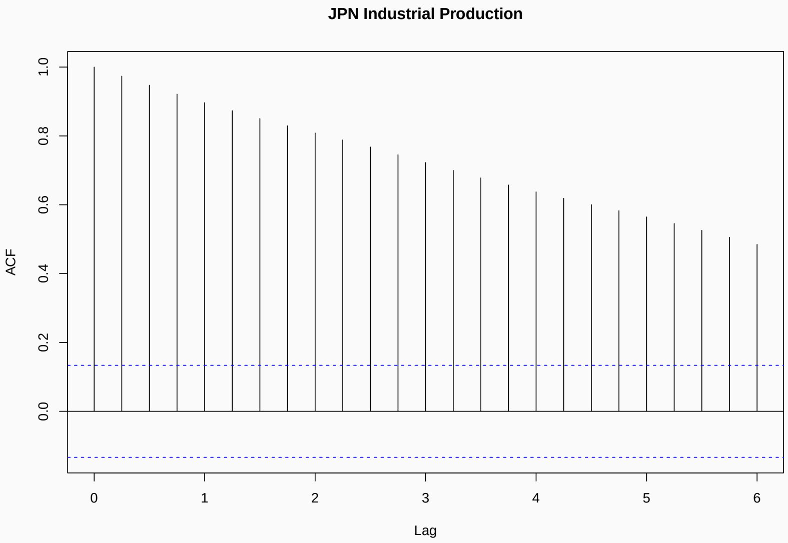
Some examples of economic time series



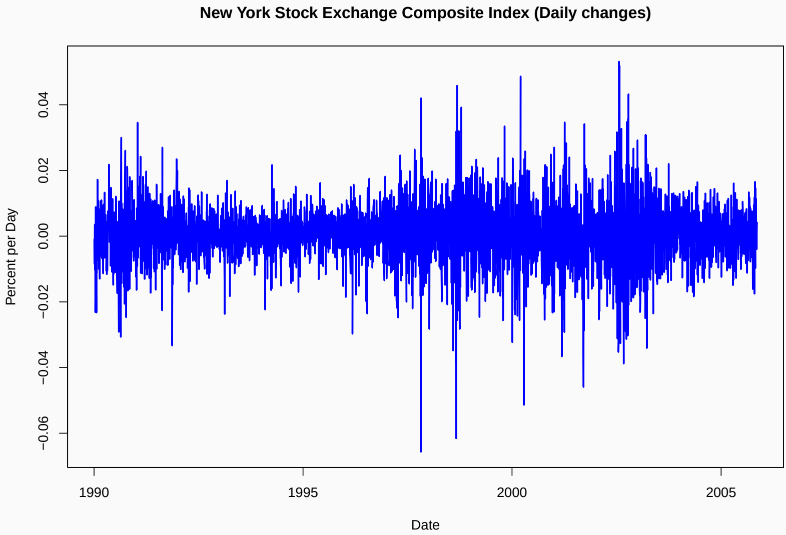
Some examples of economic time series



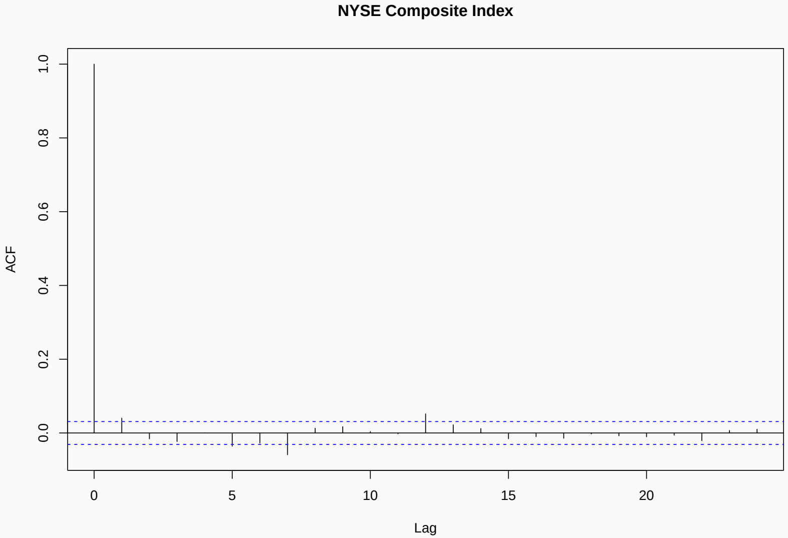
Some examples of economic time series



Some examples of economic time series



Some examples of economic time series



Central Limit Theorem

- Given a sufficiently large sample size, the sampling distribution of the sample mean will approximate a normal distribution.
- This is true regardless of the shape of the population distribution from which the sample is drawn.

Central Limit Theorem

Let $X_1, X_2, ..., X_n$ be any sequence of independent identically distributed random variables with finite expectation μ and finite variance σ^2. In particular be a random sample of size n from a population with a mean μ and variance σ^2. As n becomes large, the distribution of the sample means will approximate a normal distribution ($N(\mu, \sigma^2/n)$).

Stationarity Condition

- Consider a linear regression model (here, $AR(1)$):

$$y_t = \theta y_{t-1} + \epsilon_t, \quad t = 1, 2, \ldots, T.$$

For the OLS estimator to be consistent,

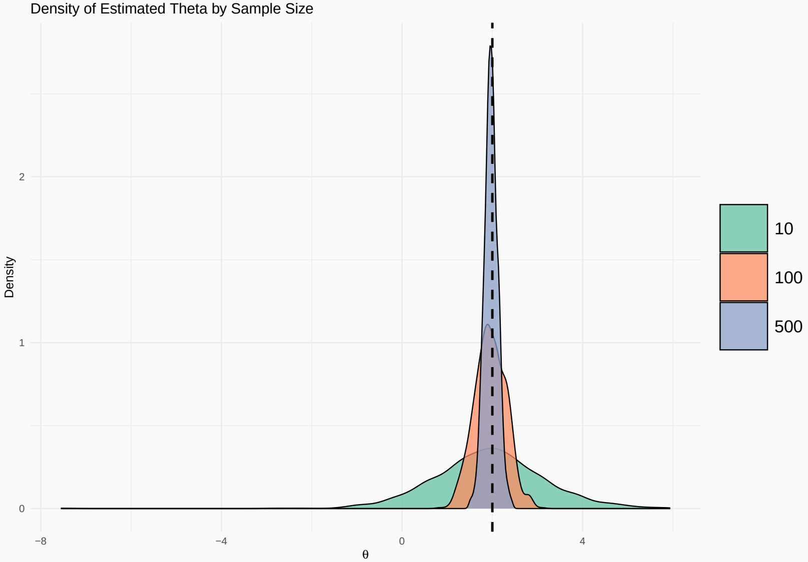
$$\hat{\theta} = \frac{\sum_{t=1}^{T} y_t y_{t-1}}{\sum_{t=1}^{T} y_{t-1}^2} \xrightarrow{p} \theta_0,$$

we typically rely on a law of a large numbers, such that sample moments converge to population moments:

$$\frac{1}{T}\sum_{t=1}^{T} y_{t-1}^2 \xrightarrow{p} E\left(y_{t-1}^2\right) \quad \text{or} \quad \frac{1}{T}\sum_{t=1}^{T} y_t y_{t-1} \xrightarrow{p} E\left(y_t y_{t-1}\right).$$

- This holds if y_t is stationarity andweakly dependent.
- For each model we will find the stationarity condition.

if y_t is stationarity and weakly dependent

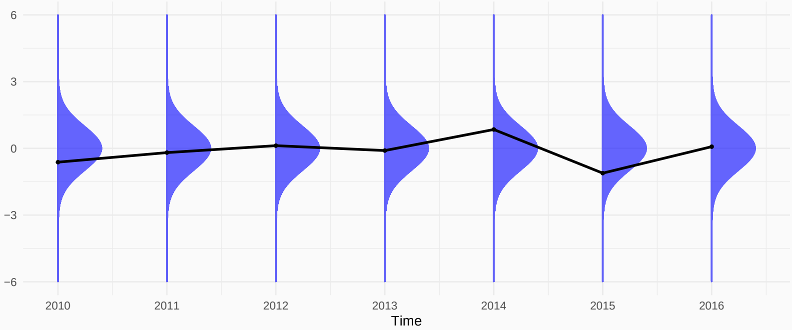


Strict Stationarity

A time series, $\{y_t\}_{t=1}^{T}$, is called strictly stationary if the distributions of

$$(y_t, y_{t+1}, \ldots, y_{t+s}) \text{ and } (y_{t+h}, y_{t+1+h}, \ldots, y_{t+s+h})$$

are the same for all h. The distribution of y_t does not depend on t.



Strict Stationarity

- If we could rerun history M times:

Stochastic process:	$y_1,$	$y_2,$	$\ldots,$	$y_t,$	$\ldots,$	y_T
Realization 1:	$y_1^1,$	$y_2^1,$	$\ldots,$	$y_t^1,$	$\ldots,$	y_T^1
	$\vdots$	$\vdots$		$\vdots$		$\vdots$
Realization m:	$y_1^m,$	$y_2^m,$	$\ldots,$	$y_t^m,$	$\ldots,$	y_T^m
	$\vdots$	$\vdots$		$\vdots$		$\vdots$
Realization M:	$y_1^M,$	$y_2^M,$	$\ldots,$	$y_t^M,$	$\ldots,$	y_T^M

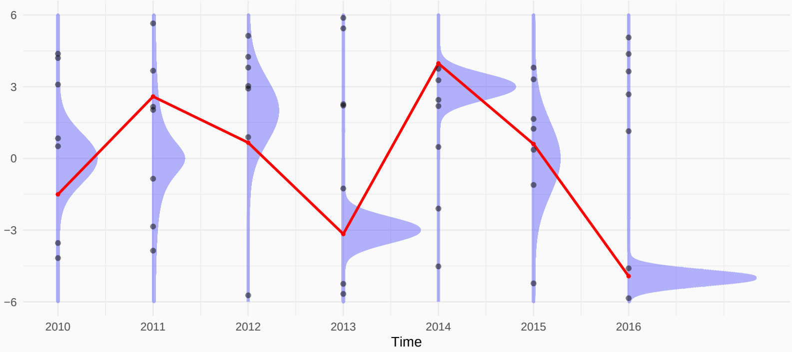
- Define the ensemble mean, $E(y_t)$. Estimated with the cross-sectional average

$$\widehat{E}(y_t) = \frac{1}{M}\sum_{m=1}^{M} y_t^{(m)}$$

Strict Stationarity

- Fundamentally different from the time average of a realized sample path

$$\bar{y}_T = \frac{1}{T}\sum_{t=1}^{T} y_t$$



Weak Stationarity

A related definition focuses on moments.

The time series is called weakly stationary if

$$
\begin{aligned}
E\left(y_t\right) &= & \mu \\
V\left(y_t\right) &= E\left(\left(y_t-\mu\right)^2\right) &= \gamma_0 \qquad \text{for } h=1,2,\ldots \\
\operatorname{Cov}\left(y_t, y_{t-h}\right) &= E\left(\left(y_t-\mu\right)\left(y_{t-h}-\mu\right)\right) &= \gamma_h
\end{aligned}
$$

Where we typically assume that $\gamma_0 < \infty$.

Finally:

The time series is called weakly dependent if y_t and y_{t-h} becomes approximately independent for $h \to \infty$.

Stationarity and weak dependence are different properties. In many cases, however, implied by the same restrictions.

Moving Average

- MA is one of the most basic time series models.
- Moving average(MA) processes are constructed from sequences of independent, identically distributed random variables ϵ_t with $E(\epsilon_t) = 0$ and $Var(\epsilon_t) = \sigma^2$ (sometimes called white noise). One can think of ϵ_t as random shocks.

Moving Average

The moving average model builds the variable u out of present and lagged (past) values of the random error (shock) ϵ. Its simplest form is

$$u_t = \epsilon_t + \theta\epsilon_{t-1}$$

and it is called the 1st order moving average model and abbreviated $MA(1)$. Again the shocks ϵ_t are assumed to be white noise.

Moving Average

MA(1) processes can be extended so they have "higher-order dynamics" or more lagged terms,

$$u_t = \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2}$$

Moving Average

Here we consider the simplest case $MA(1)$.

- Consider 3 consecutive values of u, say,

$$u_{t-1} = \epsilon_{t-1} + \theta\epsilon_{t-2}$$

$$u_t = \epsilon_t + \theta\epsilon_{t-1}$$

$$u_{t+1} = \epsilon_{t+1} + \theta\epsilon_t$$

Adjacent u's are correlated because they both depend on a common shock, e.g., u_t and u_{t+1} both depend on ϵ_t.

Moving Average

The first and second moments (means and variance/covariance) can be worked out:

- Expected value of u_t:

$$E(u_t) = E(\epsilon_t + \theta\epsilon_{t-1}) = 0$$

- Variance of u_t:

$$Var(u_t) = E(u_t^2) = E((\epsilon_t + \theta\epsilon_{t-1})^2) = (1 + \theta^2)\sigma^2$$

- Covariance of u_t with u_{t-1}:

$$E(u_t u_{t-1}) = E((\epsilon_t + \theta\epsilon_{t-1})(\epsilon_{t-1} + \theta\epsilon_{t-2})) = \theta\sigma^2$$

- Covariance of u_t with u_{t-k}, where $k > 1$:

$$cov(u_t, u_{t-k}) = E(u_t u_{t-k}) = 0$$

Moving Average

The $MA(1)$ is stationary in the sense that:

- The mean $E(u_t)$ is the same for all t (here it is 0).
- The variance of $E(u_t)$ is the same for all t (here it is $\sigma^2(1+\theta^2)$).
- The covariance $cov(u_t u_{t-k})$ does not depend on t but only on k; e.g., the covariance between u_1 and u_4 is the same as that between u_{1001} and u_{1004}.

Moving Average

The correlations between successive u's in a time series are called autocorrelations. The correlation between u_t and u_{t-k} is:

$$\rho_k = \frac{cov(u_t, u_{t-k})}{\sqrt{Var(u_t)Var(u_{t-k})}}$$

For $MA(1)$:

$$\rho_k = \frac{\theta\sigma^2}{\sqrt{(1+\theta^2)\sigma^2(1+\theta^2)\sigma^2}} = \frac{\theta}{1+\theta^2} \text{ if } k = \pm 1, \text{ and 0 otherwise.}$$

For the $MA(1)$, the autocorrelations die out after one period.

Summary

- Time series data is data collected for a single entity at multiple points in time.
- Time series data differ from cross section data in that observations are correlated.
- The covariance between a variable and its past realizations is called autocovariance.
- The autocovariance of a time series allows us to predict future values from past values.
- Stationarity.
- $MA(1)$.